Unit 12, Lesson 12: Solving One-Step Multiplication and Division Equations Algebraically Equation

Benchmark

MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

Learning Target

- ☐ I can solve one-step multiplication and division equations in one-variable algebraically.

12.12.1 Warm-Up: Different Problems, Same Solution

1. Write four equations whose solutions are all $x = 3$.

12.12.2 Guided Instruction: Using Inverses to Solve Equations

Inverse operations are used to solve equations by isolating a variable of interest.

When solving one-step equations using visual models, the inverse operation performed on one side of the model is also performed on the other side to keep the equation balanced.

1. With your partner, re-visit the meaning of balanced equations. Write a brief summary explaining what it means for equations to be balanced and how balance is maintained when solving. Include examples as part of your summary.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	**Inverse Operation**
Multiplication ($\times$)	
Division ($\div$)	

3. Consider the equation $7m = 21$ to complete the following.

   a. Discuss with your partner how to solve $7m = 21$.

   b. What operation is occurring between the variable m and its coefficient of 7?

   c. To isolate the variable m from its coefficient of 7, what inverse operation should be performed?

d. Work with your partner to determine the value of m.

4. Consider the equation $13 = \frac{h}{-5}$ to complete the following.

a. What mathematical operation is represented by $\frac{h}{-5}$?

b. What is the inverse operation?

c. Perform the inverse operation in the boxes below.

$$\boxed{}(13) = \left(\frac{h}{-5}\right)\boxed{}$$

d. What is the value of h?

e. Discuss with your partner how to check whether or not your solution is correct. Then, check your solution.

12.12.3 Guided Practice: Solving One-Step Multiplication and Division Equations Algebraically

1. Complete the table.

Equation	Description of How to Solve	Work	Solution
$6x = 18$	Divide both sides of the equation by 6.		
$\frac{h}{5} = 7$			
$-36 = \frac{m}{-6}$			
$^{-}32 = 8y$			
$4 = -\frac{b}{12}$			

2. With your partner, verify each solution is correct using substitution.

12.12.4 Your Turn!

Solve each equation algebraically. Show your work.

1. $\frac{a}{4} = 10$	2. $-11 = \frac{x}{8}$
3. $-12p = -84$	4. $-42 = 6x$
5. $18c = 72$	6. $\frac{r}{6} = 36$

12.12.5 Wrap-Up: Error Analysis

1. While checking her homework from last night, Lola noticed she got the following problem incorrect. Examine Lola's work and write a note to her detailing how she can correct her mistake.

$$\begin{array}{rl} 42 = & h \div 7 \\ +7 & +\,7 \\ \hline 49 = & h \end{array}$$